

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester III)
Data Structure Practical

Practical – V

Roll No.: S021	Name: LODHI IQRA
Class: SYBSc IT	Subject: DATA STRUCTURE
Date of Assignment: 22/07/2026	Date/Time of Submission: 22/07/2026

Binary Search Tree: Write a program to:

a. Create a binary search tree.

CODE:

```
class TreeNode:
    def __init__(self, data):
        self.data = data
        self.left = None
        self.right = None

root = TreeNode('R')
nodeA = TreeNode('A')
nodeB = TreeNode('B')
nodeC = TreeNode('C')
nodeD = TreeNode('D')
nodeE = TreeNode('E')
nodeF = TreeNode('F')
nodeG = TreeNode('G')

root.left = nodeA
root.right = nodeB

nodeA.left = nodeC
nodeA.right = nodeD

nodeB.left = nodeE
nodeB.right = nodeF

nodeF.left = nodeG

print("root.right.left.data:", root.right.left.data)
print("IQRA S021")
```

OUTPUT:

```
=====
root.right.left.data: E
IQRA S021
```

b. Insert nodes into a binary search tree.

CODE:

```
File Edit Format Run Options Window Help

class TreeNode:
    def __init__(self,data):
        self.data=data
        self.left=None
        self.right=None

root = TreeNode('R')
nodeA = TreeNode('A')
nodeB = TreeNode('B')
nodeC = TreeNode('C')
nodeD = TreeNode('D')
nodeE = TreeNode('E')
nodeF = TreeNode('F')
nodeG = TreeNode('G')
nodeH = TreeNode('H')

root.left=nodeA
root.right=nodeB

nodeA.left=nodeC
nodeA.right=nodeD

nodeB.left=nodeE
nodeB.right=nodeF

nodeF.left=nodeG
nodeH.right=nodeH

print("root.right.left.data:",root.right.left.data)
print("IQRA S021")
```

OUTPUT:

```
===== |
root.right.left.data: E
IQRA S021
```

c. Search for a node in a binary search tree.

CODE:

```

class TreeNode:
    def __init__(self, data):
        self.data = data
        self.left = None
        self.right = None

root = TreeNode('R')
nodeA = TreeNode('A')
nodeB = TreeNode('B')
nodeC = TreeNode('C')
nodeD = TreeNode('D')
nodeE = TreeNode('E')
nodeF = TreeNode('F')
nodeG = TreeNode('G')
nodeH = TreeNode('H')

root.left = nodeA
root.right = nodeB

nodeA.left = nodeC
nodeA.right = nodeD

nodeB.left = nodeE
nodeB.right = nodeF

nodeF.left = nodeG
nodeH.right = nodeH

print("root.right.left.data:", root.right.left.data)

def search(root, key):
    if root is None:
        return False

    print("Checking: ", root.data)

    if root.data == key:
        return True

    return search(root.left, key) or search(root.right, key)

key = input("Enter element to search: ")

if search(root, key):
    print("Element Found")
else:
    print("Element not Found")
print("IQRA S021")

```

OUTPUT:

```
root.right.left.data: E
Enter element to search: A
Checking: R
Checking: A
Element Found
IQRA S021
```

```
root.right.left.data: E
Enter element to search: I
Checking: R
Checking: A
Checking: C
Checking: D
Checking: B
Checking: E
Checking: F
Checking: G
Element not Found
IQRA S021
```